ML Model Evaluation & Explainability Report

Generated on 2026-09-23 13:00:36 | Session ID: exp_test

1. Dataset & Problem Detection Summary

Dataset File:	loan_data.csv	**Total Rows:**	5000
Target Column:	Loan_Status	**Total Columns:**	16
Problem Type:	Classification	**Test Split:**	20%

2. Preprocessing & Validation Protocol

Numerical Imputer	Median	**Categorical Encoding**	Onehot
Categorical Imputer	Most_frequent	**Feature Scaling**	Standard
Random State	42	**CV Protocol**	5 Folds

3. Model Performance Comparison

Rank	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Train Time (s)
1	Random Forest	0.7500	0.5000	0.1000	0.1667	0.8000	4.48

4. Recommended Champion Model

Recommended Model: Random Forest
Criterion: Highest F1 Score (0.1667).
Rationale: High accuracy and low variance

5. Global Feature Importance (SHAP)

Feature	Mean Absolute SHAP Value	Relative Impact
Credit_Score	0.1053	100.0%
Employment_Type_Salaried	0.0325	30.9%
Income	0.0321	30.5%
Employment_Type_Unemployed	0.0245	23.3%
Age	0.0229	21.7%
Existing_Loans	0.0213	20.2%
Experience	0.0167	15.9%
Savings_Balance	0.0135	12.8%